Regular Expressions

The time is finally upon us! I've been promising/threatening these for most of the chapter and decided to keep them for the end of our foray into all things text: saving the best until last. A regular expression (also known as *regex*, or *regexp*) is a sequence of characters and metacharacters used to define a pattern, which is then applied to text (often referred to as the *sample*): the regex engine will search for the pattern, and if found, it can then return portions of the text in or around the matching areas, or even replace the matching portions.

Regular expressions have their roots in the work of the mathematician Stephen Kleene, a student of Alonzo Church (see Chapter 3, "Functions"), who devised a notation system for describing (regular) events in neural networks in a 1951 Project RAND paper. (The * metacharacter – indicating zero or more repetitions and found in many guises of pattern matching systems in computing – is named the *Kleene star*.) Regular expressions as we would recognize them today were introduced by Ken Thompson (designer and implementer of technologies you might have heard of, such as Unix, C, Golang, UTF-8… quite an incredible list) in the late 1960s as text search functionality in the QED text editor. The syntax of regular expressions has varied over the decades but reached something of an apotheosis with the Perl programming language (of which I have many fond memories). PHP, like many other languages and systems, has adopted Perl's regular expression syntax (which is given the name PCRE, for "Perl-Compatible Regular Expressions"), and all of PHP's core PCRE functions are prefixed with `preg_` – such as `preg_match()`.

Before we get started, a word of caution. The study of regular expressions is an entire branch of programming and theory in its own right. This section is decidedly *not* a comprehensive dive; the intention is to just get you started. There are, happily, vast amounts of resources out there on the web and many excellent books devoted exclusively to the topic.

Basic Syntax

A Perl (and therefore PHP) regular expression is simply a string containing literal characters, and/or metacharacters, with a delimiter character (a forward slash / by convention) at the very beginning and end of the pattern. After the final delimiter there are some additional characters that can be used to modify behavior (such as case sensitivity). They `"/look like this/"`, `"/or (this)/i"`, or even `"/like [Tt][Hh][Is][Ss]/"`.